

Deepika Menon

Embedded Systems & IoT Firmware Engineer • 4 Years Experience • MERIQ Index: 69% (Developing)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: I2C/SPI/UART, ARM Cortex, MQTT, PCB Debugging, FreeRTOS, Python

Verified Certification: Embedded Linux Developer Certificate

PROFESSIONAL WORK EXPERIENCE

Embedded Systems & IoT Firmware Engineer — *Texas Instruments* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core I2C/SPI/UART service, reducing latency by 34%.
- Led a squad of 6 engineers delivering scalable microservices processing 11M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 64%.

Associate Embedded Systems & IoT Firmware Engineer — *STMicroelectronics* (2022 - 2023)

- Designed and maintained ARM Cortex data endpoints with 98% SLA uptime.
- Optimized database indexing and caching layers using MQTT saving \$29k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [I2C/SPI/UART, ARM Cortex, MQTT, PCB Debugging]

Engineered end-to-end production solution leveraging I2C/SPI/UART and ARM Cortex with full test automation.

Automotive CAN-Bus Telemetry Gateway with Embedded Linux and Secure Boot [MQTT, PCB Debugging, FreeRTOS, Python]

Architected high-resilience system integrating MQTT with 99.9% uptime.

EDUCATION & CREDENTIALS

M.Tech in Data Science & Artificial Intelligence

Harvard University — 2018 - 2022 (Score: 8.18 / 10.0)